JUNOH JUNG

Postdoctoral Appointee | Mathematics and Computer Science Division | Argonne National Laboratory
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RESEARCH PROFILE

Computational scientist developing physics-based and machine-learning methods to improve simulation accuracy and efficiency and to predict and control fluid flows. My research combines scientific machine learning, differentiable simulation, and high-performance computing to solve scientific problems.

EDUCATION

Ph.D., Mechanical Engineering, University of Michigan 2019–2024
Dissertation: Resolvent-based Estimation and Control of Aerodynamic Flows. Advisor: Aaron Towne.

M.S., Aerospace Engineering, RWTH Aachen University 2014–2018
Thesis: Development of a fully coupled zonal RANS/LES method for the simulation of a turbulent backward-facing step flow. Advisor: Wolfgang Schroeder.

B.S., Mechanical and System Design Engineering, Hongik University 2008–2014
Graduating rank: 1 of 166 in the department.

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Postdoctoral Appointee, Argonne National Laboratory Nov 2024–present
Mathematics and Computer Science Division. Supervisor: Emil Constantinescu; Mentor: Bethany Lusch.
Developing differentiable and hybrid physics–ML methods for spectral-element and finite-element flow simulations, including weak-form corrections and agentic multi-fidelity workflows. Lead Diff-NekRS; the distributed differentiable training framework has been tested at up to 1,020 MPI ranks on Aurora.

Graduate Student Research Assistant, University of Michigan Aug 2019–Aug 2024
Mechanical Engineering. Advisor: Aaron Towne.
Developed resolvent-based methods for estimation and control of aerodynamic flows, including laminar airfoil wakes and turbulent wakes.

Givens Associate, Argonne National Laboratory 2023
Mathematics and Computer Science Division. Supervisor: David Lenz; PI: Thomas Peterka.
Researched physics-informed B-spline methods for flow-data reconstruction.

Master's Thesis Researcher, RWTH Aachen University 2017–2018
Aerodynamisches Institut Aachen (AIA). Advisor: Wolfgang Schroeder.
Developed a fully coupled zonal RANS/LES method for turbulent backward-facing-step flow.

Visiting Scholar, Georgia Institute of Technology 2017
Department of Aerospace Engineering. Supervisor: Dimitri Mavris.

RESEARCH FUNDING AND COMPUTING ALLOCATIONS

Argonne Laboratory Directed Research and Development (LDRD) Apr–Sep 2026
Sole Principal Investigator | Research award: $25,000.
Project: Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics–Machine Learning Simulations.

ALCF Director's Discretionary Allocation Program 2025–present
Argonne Leadership Computing Facility (ALCF).
Sole Principal Investigator | Computing allocation: **45,000 node-hours per six-month period.**
Project: Differentiable Hybrid Physics Machine Learning Simulations.

PUBLICATIONS AND MANUSCRIPTS

[*] *Corresponding author.*

Journal Manuscripts in Preparation or Under Review

J1. **Jung, J.**, Balin, R., Lusch, B., Constantinescu, E. Diff-NekRS: Differentiable hybrid modeling for spectral-element simulations (journal version of preprint arXiv:2609.23208).

J2. **Jung, J.**, Constantinescu, E. Structure-preserving neural variational correction operators for hybrid finite-element machine-learning simulations (journal version of ASME FEDSM 2026).

J3. **Jung, J.**, Balin, R., Constantinescu, E. Agent-orchestrated multi-fidelity workflow for a stable hybrid physics–machine learning simulation. (in preparation)

Peer-Reviewed Journal Articles

J4. **Jung, J.**[*], Lenz, D., Constantinescu, E., Peterka, T. (2027). A physics-informed B-spline reconstruction of flow data. *Computer Methods in Applied Mechanics and Engineering*, 463 (Part B), 119436. doi: 10.1016/j.cma.2026.119436.

J5. **Jung, J.**[*], Towne, A. (2026). Resolvent-based estimation of a turbulent wake. *Journal of Fluid Mechanics*, 1033, A22. doi: 10.1017/jfm.2026.11444.

J6. **Jung, J.**[*], Bhagwat, R., Towne, A. (2025). Resolvent-based estimation and control of a laminar airfoil wake. *Journal of Fluid Mechanics*, 1016, A41. doi: 10.1017/jfm.2025.10423 (JFM Emerging Scholar Best Paper Prize nomination).

J7. Farghadan, A., **Jung, J.**, Bhagwat, R., Towne, A.[*] (2024). Efficient harmonic resolvent analysis via time stepping. *Theoretical and Computational Fluid Dynamics*, 38, 331–353. doi: 10.1007/s00162-024-00694-1.

J8. Martini, E.[*], **Jung, J.**, Cavalieri, A. V. G., Jordan, P., Towne, A. (2022). Resolvent-based tools for optimal estimation and control via the Wiener–Hopf formalism. *Journal of Fluid Mechanics*, 937, A19. doi: 10.1017/jfm.2022.102.

Conference Manuscripts and Forthcoming Papers

C1. **Jung, J.**, Constantinescu, E., Balin, R. (2027). Agent-orchestrated multi-fidelity workflow for stable hybrid physics–machine learning simulations. *AIAA SciTech Forum* (accepted).

C2. **Jung, J.**[*], Balin, R., Lusch, B., Constantinescu, E. (2026). Diff-NekRS: A scalable differentiable framework for multi-timestep solver-in-the-loop training. *Submitted.*

Conference Papers and Proceedings

C3. **Jung, J.**, Constantinescu, E., Balin, R., Lusch, B. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. *AIAA Aviation Forum* (Best Paper Award nomination).

C4. **Jung, J.**, Constantinescu, E. (2026). Learning differentiable weak-form corrections to accelerate finite element simulations. *ASME FEDSM 2026*.

C5. Zhou, Y., Towne, A., **Jung, J.**, Bhagwat, R., Martini, E., Jordan, P., Audiffred, D., Maia, I., Cavalieri, A. V. G. (2026). Toward resolvent-based estimation and control of wavepackets in supersonic turbulent jets. *AIAA SciTech Forum*.

C6. Towne, A., Bhagwat, R., Zhou, Y., **Jung, J.**, Martini, E., Jordan, P., Audiffred, D. B. S., Maia, I., Cavalieri, A. V. G. (2024). Resolvent-based estimation of wavepackets in turbulent jets. *AIAA/CEAS 2024*.

C7. Towne, A., **Jung, J.**, Bhagwat, R., Zhou, Y., Paulson, J. (2024). Resolvent-based estimation and control of wakes and jets. *13th International Symposium on Turbulence and Shear Flow Phenomena*.

C8. **Jung, J.**, Towne, A. (2024). Toward turbulent wake estimation using a resolvent-based approach. *AIAA SciTech Forum*.

C9. **Jung, J.**, Bhagwat, R., Towne, A. (2023). Resolvent-based estimation of laminar flow around an airfoil. *AIAA SciTech Forum*.

HONORS, FELLOWSHIPS, AND RECOGNITION

ALCF Postdoctoral Fellowship, Argonne National Laboratory Nov 2024

Selected Participant, Argonne Training Program on Extreme-Scale Computing Aug 2023
ATPESC; funded by the U.S. Department of Energy. Selected from about 500 applicants.

Rackham Travel Grant, University of Michigan Winter 2022; Winter 2023

Mechanical Engineering Fellowship, University of Michigan Winter 2020

Merit-based Academic Scholarships, Hongik University 2009–2014
Academic excellence scholarships, including full-tuition awards.

TEACHING AND MENTORING

STARS High School Research Program at Argonne National Laboratory 2026–2027
Mentoring high-school students who are interested in engineering for research projects (upcoming)

Mentor and Participant, Argonne GPU Hackathons 2025–2026
Mentored NekRS/NekRS-ML teams on GPU porting and performance of scientific applications.

Informal Mentor, Yuhao Zhou (Ph.D. student) 2022–2025

Graduate Student Instructor, University of Michigan 2024
Introduction to Fluid Mechanics (ME 320). Supported 101 students across two course sections (ME 320-001 and ME 320-002) through office hours, recitations, quiz review, and exam grading.

TECHNICAL EXPERTISE

Scientific machine learning: Differentiable simulation (Diff-NekRS); physics-informed B-spline reconstruction; weak-form corrections; deep learning (CNNs, GNNs, U-Net, Transformers); agentic simulation workflows.

Computational fluid dynamics and reduced-order methods: DNS, LES, RANS; spectral-element and finite-element methods; flow estimation and control; DMD, POD, SPOD, and resolvent analysis.

Programming and HPC: C/C++, Python, Fortran; MPI (OpenMPI, MPICH); GPU and heterogeneous computing; PETSc, FFTW, LAPACK.

Tools and computing environments: Git/GitHub, Ubuntu, Bash; DOE, DoD, and university supercomputing environments; Tecplot, ParaView, VisIt; UG-NX and AutoCAD.

PROFESSIONAL SERVICE AND LEADERSHIP

Session Chair, APS Division of Fluid Dynamics Meeting 2026 (upcoming)

Proposal Reviewer, APEX Program at ALCF, Argonne National Laboratory 2026

Journal Referee 2026
Journal of Fluid Mechanics, Journal of Computational Physics, Computer Methods in Applied Mechanics and Engineering, and Scientific Reports.

Abstract Reviewer, AIAA SciTech and Aviation 2024–2026
Reviewed fluid dynamics submissions.

Elected Local President Oct 2014–Oct 2016
Korean-German Scientists and Engineers Association. Elected to represent the Aachen, Cologne, and Bonn region.

Professional Memberships

Association for Computing Machinery (ACM), 2026; American Institute of Chemical Engineers (AIChE), 2026; American Society of Mechanical Engineers (ASME), 2026.

American Institute of Aeronautics and Astronautics (AIAA), 2022–2026; American Physical Society (APS), 2020–2026.

COLLABORATIVE PROPOSAL DEVELOPMENT

Key Personnel, Genesis Mission Phase 1 Proposal Apr 2026
Led by Arizona State University; not selected. Project: AI-Enabled Design of Electrolyte Wetting Processes in Thick Porous Battery Electrodes. Focus: AI-enabled scientific workflows and adaptive learning methods for energy applications.

ADDITIONAL PROFESSIONAL ACTIVITIES

Invited Participant, Hyundai Motors Vision Conference Aug 2022
Invited to South Korea for discussions with UAM engineers and factory visits.

Sergeant, Republic of Korea Army Feb 2010–Dec 2011

INVITED TALKS

1. **Jung, J.** (2027). From weak-form correction to Agent-orchestrated workflow for hybrid physics-ML framework, SIAM CSE27. Minisymposium: "Recent Advances in Galerkin Methods for Physical Systems" (forthcoming).
2. **Jung, J.** (2024). Estimation and Control of Aerodynamic Flows Using a Resolvent-Based Approach, ALCF.

SELECTED TALKS AND PRESENTATIONS

1. **Jung, J.**, Constantinescu, E., Balin, R. (2027). Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics–Machine Learning Simulations. AIAA SciTech Forum (forthcoming).
2. **Jung, J.**, Balin, R., Lusch, B., Constantinescu, E. (2026). Diff-NekRS: Bridging Physics-Based Simulation and Machine Learning. APS Division of Fluid Dynamics, oral presentation (forthcoming).
3. **Jung, J.**, Constantinescu, E. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. Nek User Meeting.
4. **Jung, J.**, Constantinescu, E., Balin, R., Lusch, B. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. AIAA Aviation.
5. **Jung, J.**, Constantinescu, E. (2026). Learning Differentiable Weak-Form Corrections to Accelerate Finite Element Simulations. ASME FEDSM, oral presentation.
6. **Jung, J.**, Constantinescu, E. (2026). Differentiable Weak-Form Learning for Coarse-Mesh Incompressible Flow Simulation: Toward Chemical Engineering Applications, 18th AIChE Midwest Regional Conference, oral presentation.
7. **Jung, J.**, Constantinescu, E. (2026). Differentiable Weak Form Learning for Structure Preserving Finite Element Acceleration of Incompressible Flows. USNCTAM26, oral presentation.
8. **Jung, J.**, Constantinescu, E. (2025). Hybrid Physics-Machine Learning Framework Toward Efficient Simulation of Turbulent Flows on an Exascale Platform. APS Division of Fluid Dynamics, oral presentation.
9. **Jung, J.**, Constantinescu, E. (2025). Hybrid Physics-Machine Learning Modeling with post-update discrete corrections. Postdoctoral Research and Career Symposium, Argonne National Laboratory, poster presentation.
10. **Jung, J.**, Towne, A. (2024). Resolvent-based estimation and control of airfoil wakes. APS Division of Fluid Dynamics, oral presentation.
11. **Jung, J.**, Towne, A. (2024). Toward turbulent wake estimation using a resolvent-based approach. AIAA SciTech Forum.
12. **Jung, J.**, Towne, A. (2023). Resolvent-based estimation for turbulent wakes. APS Division of Fluid Dynamics, oral presentation.
13. **Jung, J.** (2023). Resolvent-based estimation and control of flows, Michigan Fluid Mechanics Research Seminar (internal), oral presentation.
14. **Jung, J.**, Bhagwat, R., Towne, A. (2023). Resolvent-based estimation of laminar flow around an airfoil. AIAA SciTech Forum.
15. **Jung, J.**, Bhagwat, R., Towne, A. (2022). Real-time estimation and control of flow over a NACA0012 airfoil using resolvent-based tools. APS Division of Fluid Dynamics, oral presentation.
16. **Jung, J.** (2021). Resolvent-based flow estimation and control, Michigan Fluid Mechanics Research Seminar (internal), oral presentation.
17. **Jung, J.**, Martini, E., Cavalieri, A., Jordan, P., Lesshafft, L., Towne, A. (2020). Optimal resolvent-based estimation for flow control. APS Division of Fluid Dynamics, oral presentation.

COAUTHORED PRESENTATIONS — PRESENTED BY COLLABORATORS

1. Zhou, Y., Bhagwat, R., **Jung, J.**, Towne, A. (2026). Resolvent-based Estimation and Control of Wavepackets in a Supersonic Jet. APS Division of Fluid Dynamics, oral presentation (forthcoming).
2. Zhou, Y., **Jung, J.**, Bhagwat, R., Towne, A. (2025). Estimation and closed-loop control of supersonic jet noise. APS Division of Fluid Dynamics, oral presentation.
3. Constantinescu, E., Kang, S., **Jung, J.** (2025). Hybrid machine learning for multiscale PDE simulation using neural ODEs. 18th U.S. National Congress on Computational Mechanics, oral presentation.
4. Towne, A., **Jung, J.**, Bhagwat, R., Zhou, Y., Paulson, J. (2024). Resolvent-based estimation and control of wakes and jets. 13th International Symposium on Turbulence and Shear Flow Phenomena, oral presentation.
5. Bhagwat, R., **Jung, J.**, Martini, E., Schmidt, O., Jordan, P., Towne, A. (2022). Toward resolvent-based estimation and control of high-speed turbulent jets. APS Division of Fluid Dynamics, oral presentation.
6. Martini, E., **Jung, J.**, Cavalieri, A. V. G., Jordan, P., Towne, A., Lesshafft, L. (2021). Optimal control for large systems: a resolvent-based approach. XXV ICTAM Conference, oral presentation.